



GradeShift | AI

Physics-Informed Reinforcement Learning for
Real-Time Polyolefin Grade Transition Optimization

Team:
Agent Zero

Event:
HMEL i-QUEST 2026

Theme: Polymers & Advanced
Materials × Digitalization & AI/ML

Every year, HMEL's polymer complex loses ₹158 Crore to off-spec polymer during grade transitions.

65,500 tonnes/yr

Off-spec material produced during grade changes

180–280 transitions/yr

Across 4 reactor lines

₹158 Crore/yr

Annual economic loss from downgrading, reprocessing, and monomer flaring

Grade Transition

Prime Polymer

₹95/kg

Off-Spec Losses

(65,500 tonnes/yr) - ₹75/kg

Across HMEL's 2.25 MMTPA polyolefin complex, grade transitions are the single largest recurring cash drain.

The polymer is not wasted—it is downgraded by 15–30% and sold into low-value applications.

HMEL operates four fundamentally different reactor technologies—each with distinct transition physics.

Novolen® PP

Vertical stirred gas-phase

Capacity: 500 KTPA

Transition: 3.5–7.5 hrs

Off-spec: 60–160 tonnes

Spheripol® PP

Bulk loop + gas-phase

Capacity: 500 KTPA

Transition: 2.0–5.0 hrs

Off-spec: 40–120 tonnes

MarTECH® HDPE

Dual slurry loop

Capacity: 450 KTPA

Transition: 2.0–4.5 hrs

Off-spec: 35–90 tonnes

UNIPOL™ PE

Gas-phase fluidized bed

Capacity: 800 KTPA

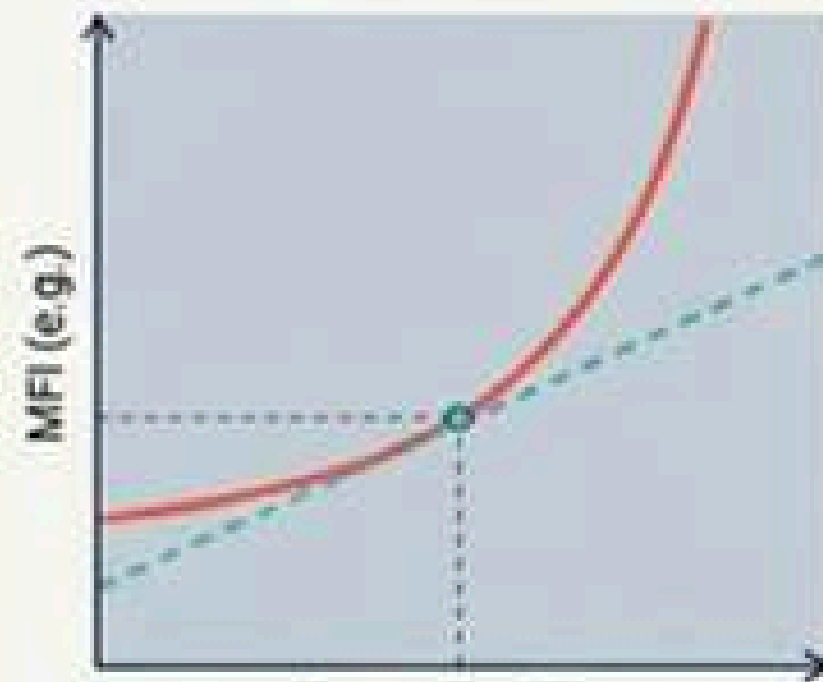
Transition: 6.0–14.0 hrs

Off-spec: 150–380 tonnes

No commercial vendor offers a unified AI optimization across all four kinetic systems. GradeShift adapts to any platform.

Three compounding blind spots make legacy Advanced Process Control (APC) fail during wide transitions.

Non-Linear Physics

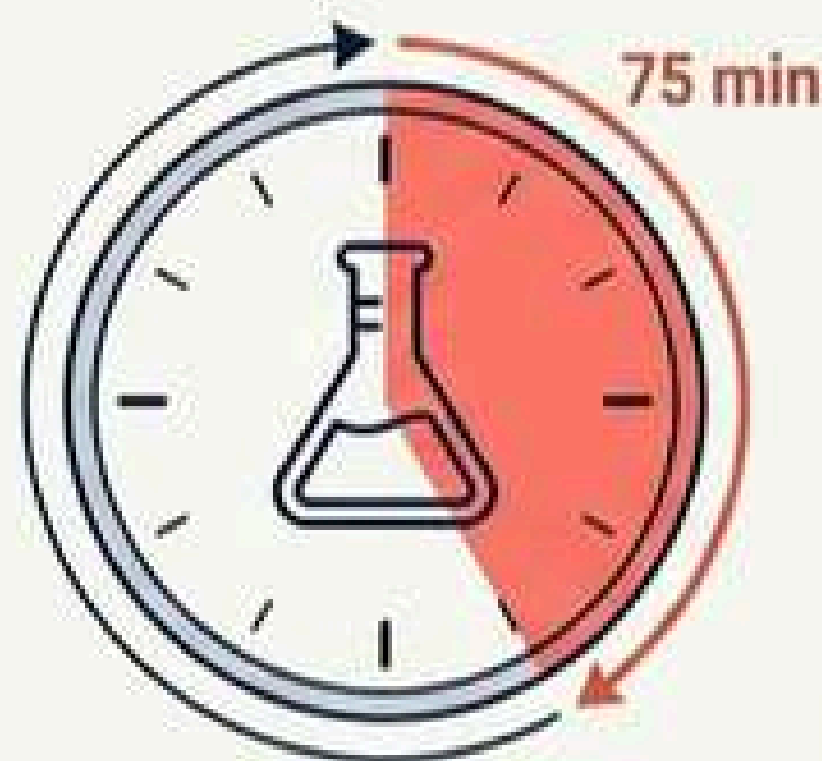


Operating Point

Linear APC models are only valid within $\pm 3\text{-}5\%$ of an operating point. During wide transitions, the model breaks.

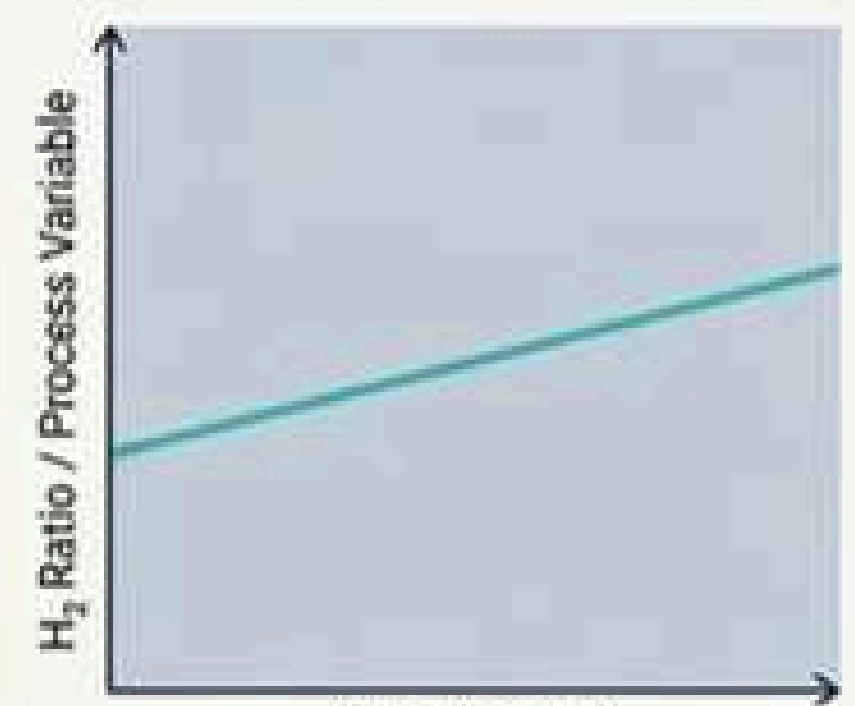
$\text{MFI} \propto \text{Mw}^{3.4}$. Service factor degrades to $<60\%$ in 18 months.

The Lab Delay



Lab MFI measurements take 35-75 minutes. At 60 T/hr, that's up to 75 tonnes produced while operators are flying blind.

Conservative Ramps

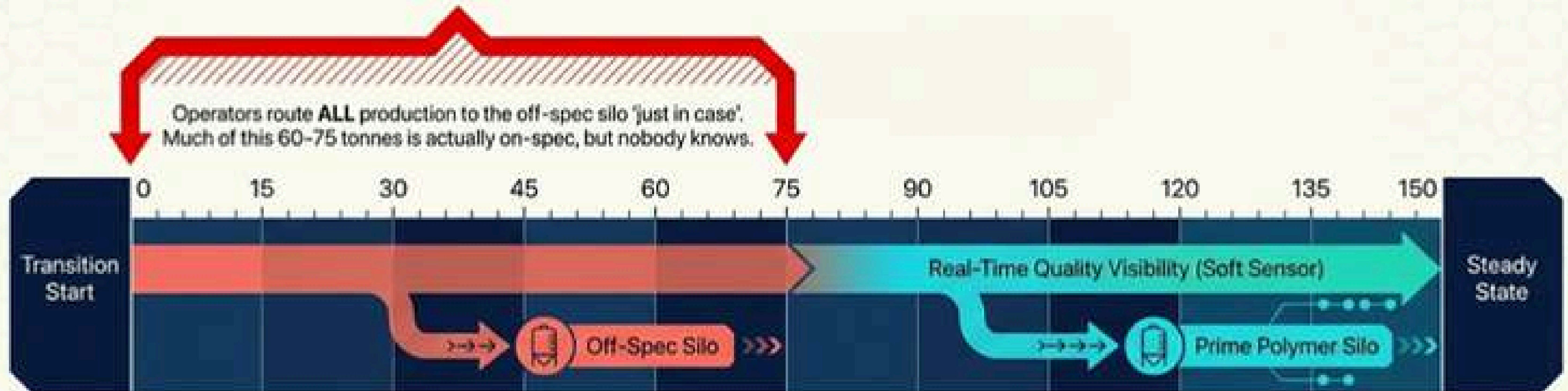


Time (hours)

DCS slowly ramps H₂ ratio linearly over 3-4 hours to avoid reactor upsets, sacrificing time for perceived stability.

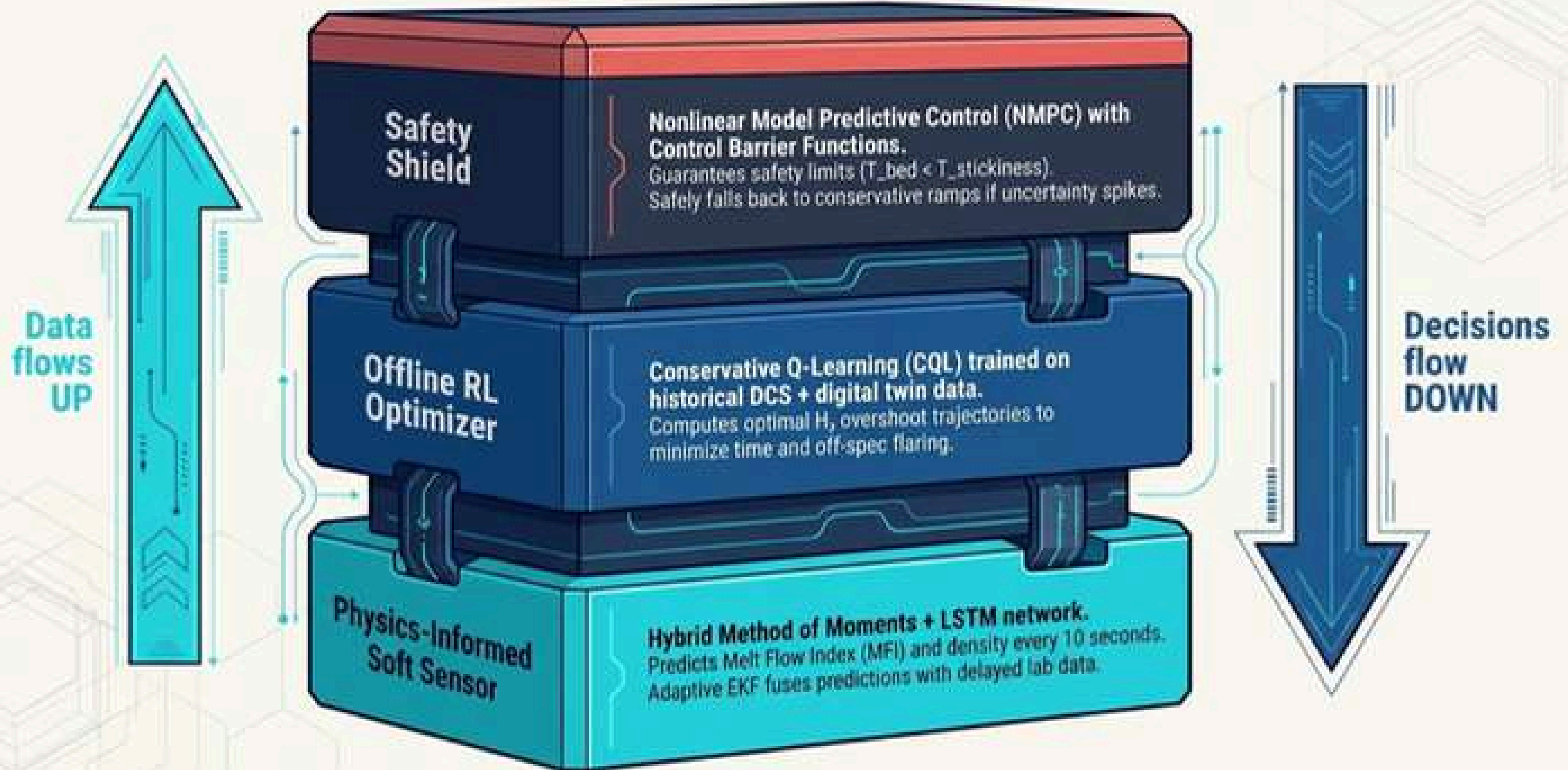
The biggest loss isn't the slow reactor kinetics—it's the 75-minute laboratory blind spot.

The 75-Minute Blind Window



Our Soft Sensor completely eliminates this 1.3-hour blind window. Just by seeing the polymer quality in real-time, every transition recovers 45-90 tonnes of prime polymer—saving ₹17 Crore/year before any reactor optimization even begins.

GradeShift Architecture: A three-tier stack marrying AI with absolute operational safety.



Governed by thermodynamics: Why plant managers can trust a physics-informed model

$$\frac{d\mu_k}{dt} = \sum \left[R_{\text{init},i} + k_{p,i}[M]\mu_{k-1,i} - k_{tr,H2,i}[H_2]\mu_{k,i} \right] - \frac{\mu_k}{\tau}$$

Method of Moments: Obeys conservation of mass and reaction kinetics for multi-site Ziegler-Natta catalysts.

$$\ln(MFI) = 88.0 - 3.40 \cdot \ln(M_w)$$

MFI Correlation: Translates molecular weight distributions accurately.

$$\frac{dMFI_{\text{bed}}}{dt} = \frac{MFI_{\text{inst}} - MFI_{\text{bed}}}{\tau}, \tau \approx 2.5 \text{ hours}$$

CSTR Washout: Accurately models bed averaging and exponential lag.

The physics backbone strictly enforces conservation of mass and energy. The ML layer only corrects for unmodeled disturbances like catalyst batch variation. It is fundamentally incapable of suggesting physically impossible states.

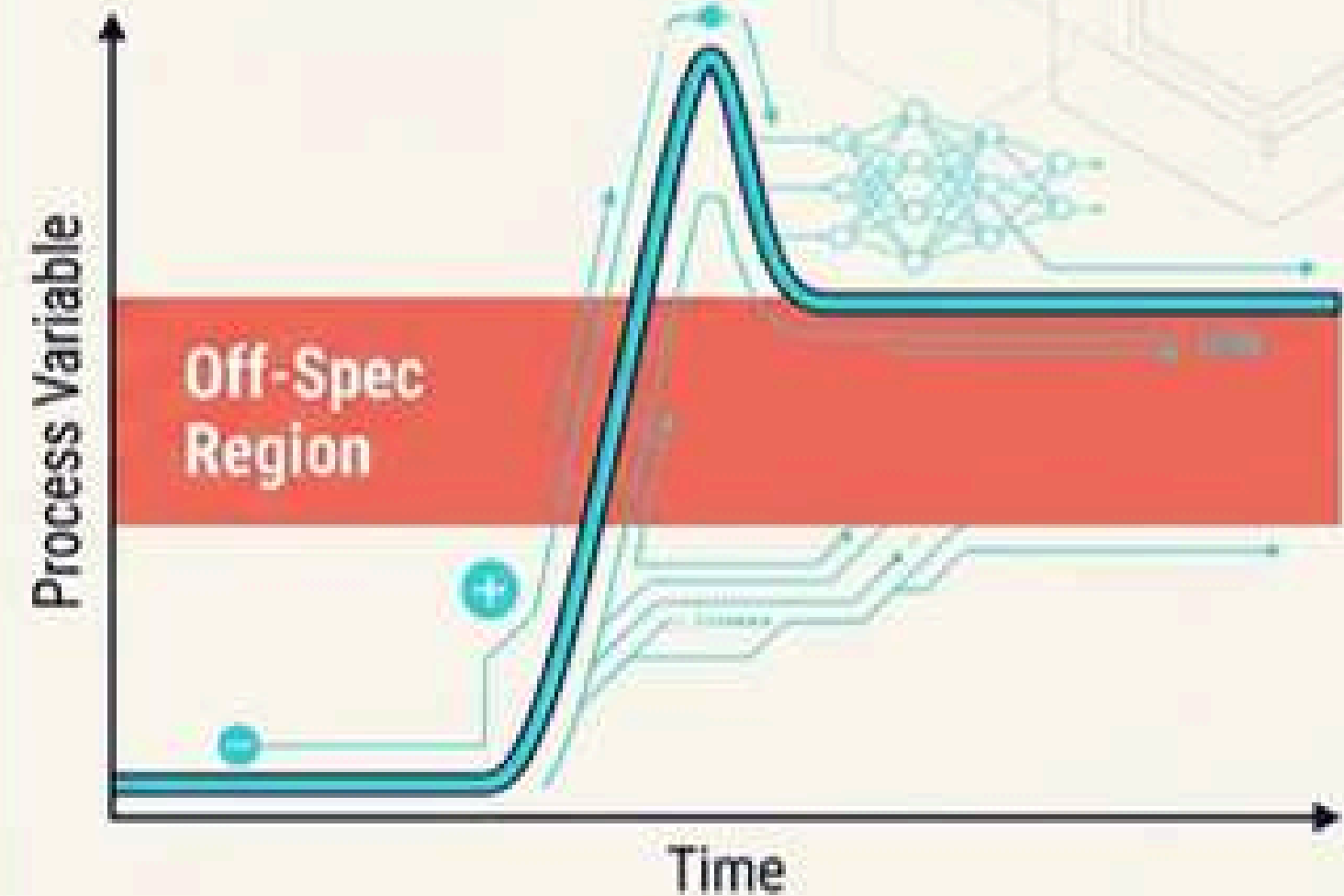
Optimal Control Theory proves that dynamic overshoot is mathematically faster than linear ramping.

Conservative Linear Ramp (Current)



Transition Time: ~8 hours
Off-Spec Material: 200 Tonnes

AI-Optimized Overshoot (GradeShift AI)



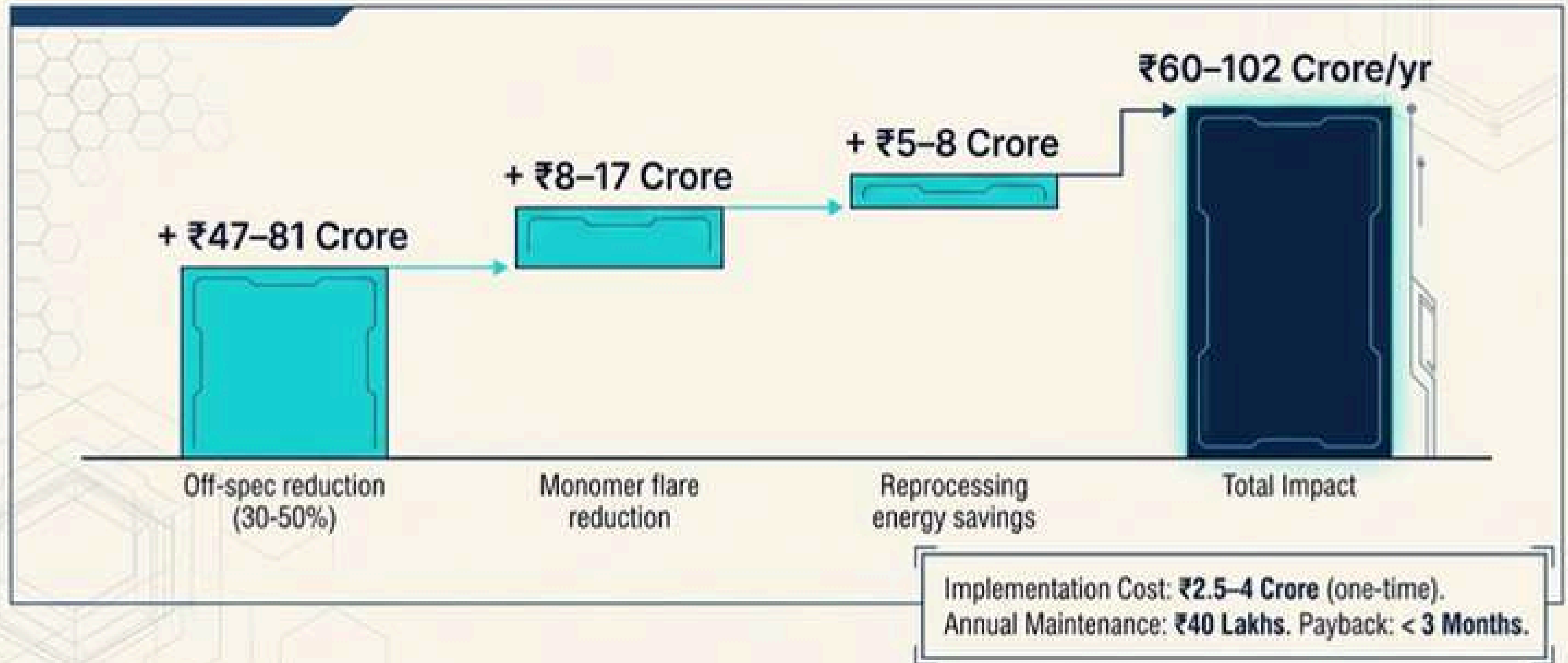
Transition Time: ~4.5 hours
Off-Spec Material: 90 Tonnes

Live POC Validation: Achieving a 44% reduction in transition time



METRIC	CONSERVATIVE RAMP	GRADESHIFT AI	IMPROVEMENT
Transition Time	8.0 hours	4.5 hours	-44%
Off-spec per transition	200 tonnes	90 tonnes	-55%
Lab delay impact	75 min blind	10 sec prediction	-99.8%
Soft sensor accuracy	—	$R^2 > 0.95$	—

₹60–102 Crore annual savings with an implementation payback period of under 3 months.



Directly supporting HMEL's 'Net Zero 2040' goals with measurable sustainability impact.



**6,000–15,000
tCO₂/yr**

Avoided emissions from
reduced monomer
flaring.



15 GWh/yr

Energy saved by eliminating
the reprocessing of
30,000+ tonnes of
downgraded polymer.



ZERO

Additional safety risk due
to the mathematical
guarantees of the Control
Barrier Functions.

The Competitive Edge: Why legacy commercial APCs cannot match **GradeShift AI**.

FEATURE CATEGORY	LEGACY APCs (Honeywell / Aspen)	GRADESHIFT AI
Model Type	Linear step-response (FIR/ARX)	Physics-informed non-linear (PINN + LSTM)
Trajectory	Fixed linear ramps	Dynamically optimal overshoot
Lab Delay	Manual bias update every 4-8 hours	10-second continuous soft sensor
Maintenance	\$450K-\$1.8M step-testing every 3-5 years	Self-adaptive (no step-testing required)
Multi-platform capability	One vendor per reactor type	Unified framework across all 4 HMEL platforms

**We don't just make the reactor faster.
We make the plant smarter.**

1

The ₹158 Crore Prize

Targeting the single largest recurring cash loss in HMEL's petrochemical operations with a sub-3-month payback.

2

Genuinely Novel AI

Integrating offline Reinforcement Learning with physical safety shields—a capability currently unavailable from any commercial APC vendor.

3

The Non-Obvious Insight

Solving the 75-minute lab delay recovers up to 90 tonnes of prime polymer per transition before reactor optimization even begins.



GradeShift AI



URL: GradeShift.AI 